Comparing Supervised Transformers and Unsupervised Diffusion Models in RNA Structure Prediction

In this review, the two papers that I will cover are titled “Ab initio prediction of RNA structure ensembles with RNAnneal” by Herron et al., and “Accurate prediction of protein–nucleic acid complexes using RoseTTAFoldNA” by Baek et al. These papers both use machine learning to predict RNA structure in the hopes of being able to generate targetable RNA drugs. In structural biology, RNA remains a challenging macromolecule for structural prediction compared to proteins. This is due to the flexibility of RNA and the many different ways it can bind and lead to downstream cellular changes. While most proteins just fold into singular, stable shapes, RNA molecules typically exist in the cell as a more dynamic set of interchanging structures. This makes RNA a particular powerful target as it’s dynamics allows for greater Shannon entropy which means that a much greater amount of information can be stored in RNA. If a proteins’ rigid structure and binding capabilities only allow for one function, RNA's interconvertibility allows for many more roles. Traditional long-standing experimental methods like X-ray crystallography and NMR often struggle to observe these metastable structures and only provide a static "average" snapshot that can miss very important functional states. Accurately predicting these structures is vital for drug discovery and understanding viral mechanisms, where RNA flexibility often dictates biological activity.

The first paper I will be talking about uses RoseTTAFold2NA, which was developed with the objective of extending high-accuracy, end-to-end structural prediction to nucleic acids and their complexes with proteins. This research was led by David Baker, who won the Nobel Prize in Chemistry in 2024 for “Computational protein design.” The researchers hypothesized that the same mechanisms used in protein folding could be adapted to learn the specific geometric constraints of RNA and DNA. Their goal was to create accurate, static 3D coordinates for multi-component molecular assemblies. The model was trained on a massive library of experimentally solved structures from the Protein Data Bank (PDB). They decided to use a supervised 36 layer three-track SE(3)-equivariant transformer (NVIDIA. (2023). SE(3)-Transformers for PyTorch) which means that the model uses regression to predict the exact 3D coordinates for every atom, this is helpful for determining long-range dependencies in the structure where elements far away interact. These models are common in structural prediction and used by the likes of Alphafold. The model is also equivariant which means that it recognizes that the molecule is the same whether it is upside down or sideways which helps predict every way molecules can fit together (Kim, S. E. (2025). Equivariant Autoregressive Models for Molecular Generation). This is also important for 3D coordinate regression. The model works by analyzing and refining the 1D sequence data, the 2D distance maps, and the 3D atomic positions. The model communicates the information between these different tasks iteratively until the model converges on a single 3D structure that minimizes the distance from the known experimental structure. RoseTTAFold2NA is evaluated using root-mean-square deviation and local distance difference test. In these tests the model will regularly score atomic-level accuracy.

This model is incredibly fast at prediction and is very accurate for well-characterized complexes that it has seen similar to its PDB data but is limited by unknown more unique sequences that don't look like PDB training data. The model is widely used around the world as it is open source (Helsem, S. A, et al. (2025)). This model led to a big improvement in the state of RNA prediction, but now that the paper is several years old, newer models have become more common.

The second paper came out in February and created and uses RNAnneal which was developed by UMD professor Pratyush Tiwary. This model differs and represents a shift in thinking about RNA structure prediction, as it addresses the conformational problem from a different angle. This model's objective is to try to move beyond single-structure snapshots of RNA, which can be deceptive due to its flexibility, by using an *ab initio* approach to identify the entire list and likelihood of conformational states an RNA molecule can adopt. The paper hypothesized that by combining generative modeling with statistical physics, a model could identify flexibility bases, traditionally only found experimentally with SHAPE probing, without requiring pre-labeled experimental data. This is unsupervised learning, compared with the first paper, which trained on PDB labels, this model uses molecular dynamics simulations. This model uses a denoising diffusion framework, which starts with noisy data that the model refines as it learns into realistic structures (NVIDIA. (2024). Denoising Diffusion NVIDIA PhysicsNeMo Framework, NVIDIA Documentation.). They use Principal Component Analysis (PCA) to capture the highest-variance geometric features, and then identify possible conformational states that it determines using physics and calculate their relative stability based on 28 thermodynamic energy maps it has generated from its learning. The model calculates interaction entropy which is how it determines the flexibility of bases and can tell which sequences of the RNA interconvert in complex structures. The RNAnneal model is measured using area under the accumulation curve where it scored a median of 0.83 in distinguishing realistic structures from decoy structures. The model is also measured using a BEDROC score which it scored a 0.40, which significantly outperformed Rosetta (0.16) and DES-AMBER (0.26), which are the top competing models. This shows the RNAnneals' ability in 'early recognition' by consistently prioritizing experimentally resolved conformations at the very top of its ranked predictions. The model is new however and the current limit of physics models struggles with complex structures of RNA such as pseudoknots and the role of magnesium ions. RNAnneal is very flexible as it has learned from physics from the ground up without training data, and is far more generalizable than RoseTTAFold2NA, but it is more computationally expensive because it generates thousands of possible structures. This model is new so not widely implemented and currently available on a UMD academic webserver. I believe this model and strategy will be the future norm of structural prediction as novel structures are more impactful and versatile while maintaining fidelity in drug targeting rather than well-categorized structures.

While both models aim to address the challenging task of RNA structure prediction, they do so through distinct machine learning philosophies. RoseTTAFold2NA represents a supervised, data-driven approach, while RNAnneal utilizes an unsupervised, physics-based approach. RoseTTAFold2NA has its function as a specific structure tool similar to past protein structure prediction models and employs supervised learning and regression via SE(3)-equivariant transformers to predict the precise 3D coordinates of stable molecular complexes based on thousands of labeled examples from the Protein Data Bank. In contrast, RNAnneal addresses the inherent flexibility and metastability of RNA by treating the problem as a dynamic challenge. It utilizes unsupervised learning and a denoising diffusion framework to explore the energetic landscape of a molecule without requiring pre-labeled experimental data. RoseTTAFold2NA excels at providing immediate atomic-level precision for well-characterized systems, and RNAnneal provides a more generalizable path forward for drug discovery by identifying flexibility and hidden conformational states through its novel Interaction Entropy metric.

Citations

1. **Baek, M., et al. (2024).** Accurate prediction of protein-nucleic acid complexes using RoseTTAFoldNA. *Nature Methods*. https://doi.org/10.1038/s41592-023-02086-5
2. **Herron, L., et al. (2026).** Ab initio prediction of RNA structure ensembles with RNAnneal. *bioRxiv*. https://doi.org/10.64898/2026.02.01.703098
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